

Suggested FYP topics for 2026-27

by

Professor Xiaojun Lin

Email: xjlin@ie.cuhk.edu.hk

Office: SHB708

Project code :	XJL1
Project title :	Control and Communication for Multi-UAV Coordination
No. of students :	2
Description of project :	The low-altitude economy, where UAVs (unmanned aerial vehicles) help carrying out delivery, monitoring, and rescuing missions that are previously impossible, are receiving more and more interest in both the mainland and Hong Kong. As the scale of low-altitude economy continues to grow, UAV operations will gradually shift from single-UAV operation to multi-UAV coordination. However, the onboard sensing and computation capabilities of UAVs are often limited. Thus, communication with the ground stations with higher computation power could greatly enhance the coordination of multiple UAVs. In this project, we will study wireless communication and control for multi-UAV coordination through simulations. We will first simulate the operation of multiple UAVs on the Gazebo simulator, and experiment with some multi-UAV control algorithms, assuming perfect communications. Then, we will study how communication imperfection (e.g., delay and loss) will affect the UAV operation efficiency. The students on the project are expected to have good programming skills and a strong mathematical background. Some knowledge of feedback control and wireless networks will be helpful.
Industrial collaboration :	No.

Project code :	XJL2
Project title :	Exploring the use of caching in ML-based recommendation systems
No. of students :	2
Description of project :	Modern online services (such as TikTok, Xiaohongshu) are driven by sophisticated recommendation systems that can produce highly accurate and responsive product recommendations to users. Increasingly, this type of recommendation system is backed by ML methods, such as deep networks. However, implementing such recommendation systems can incur significant computation and communication resources, since it needs to fuse and process information from many sources in real-time. Caching can be an important tool to reduce the amount of computation and communication. In this project, we will study the performance of

	different caching algorithms in ML-based recommendation systems, understand their complexity and limitations, and explore potential improvements. The students on the project are expected to have strong programming skills and a strong foundation in mathematics (in particular probability). Some background knowledge of neural networks, recommendation systems, and network caching would be helpful.
Industrial collaboration :	No.

Project code :	XJL3
Project title :	Student-initiated projects
No. of students :	2
Description of project :	Potential topics can be related to computer cluster scheduling for LLM serving, low-latency 5G networks, edge networks, etc. Students interested in this option are encouraged to discuss it with Prof. Lin in advance.
Industrial collaboration :	No.

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